# Hook and loop counts

How Velcro Works

Name _____ Date _____

Show your thinking with numbers, marks, or words.

**1.** The engineer counts 48 hooks on one burr and 84 hooks on another. Put >, <, or = in the circle. What is the greater count (or the shared count if equal)?

48 ○ 84

Show your thinking.

**2.** Little Wonderer counts 63 hooks in one tiny square and 36 loops in another. Put >, <, or = in the circle. What is the greater count (or the shared count if equal)?

63 ○ 36

Show your thinking.

**3.** The big one counts 55 loops on one patch and 55 loops on a second patch. Put >, <, or = in the circle. What is the greater count (or the shared count if equal)?

55 ○ 55

Show your thinking.

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**4.** Big Wonderer peels the shoe strap open 27 times. What is 27 rounded to the nearest ten?

27 → nearest ten: ___

Show your thinking.

**5.** The bag hangs on the strip for 74 seconds before the little one peels it. What is 74 rounded to the nearest ten?

74 → nearest ten: ___

Show your thinking.

**6.** A comb lifts 45 bits of lint out of an old strip. What is 45 rounded to the nearest ten?

45 → nearest ten: ___

Show your thinking.

WONDER DAILY

# How Velcro Works



## A strap that rips

A shoe strap rips open, then presses shut and holds. Its plain name is hook and loop: one rough side, one soft.





## A burr on a walk

In 1941, burrs stuck to an engineer's socks and his dog. Under a microscope, each spine ended in a tiny hook.





## Hooks meet loops

Stiff hooks on one side catch soft loops on the other. One hook is weak, but hundreds together hold strong.





## Pull or peel?

A sideways pull is hard, because every hook pulls at once. A peel is easy: only one thin line lets go.





## A bag that hangs

A hanging bag pulls along the strip, so every hook shares the weight. Peel one corner, and the bag slides off.





## When grip fades

Nylon loops were heated, then snipped into hooks. Lint clogs old hooks, but a fine comb can lift it out.



## Make a little book



**1** Fold it into 8 boxes. Open it flat.



**2** Fold short end to short end. Cut the thick line. Open it.



**3** Fold the long way, pictures out. Push the ends in.




**4** Fold into a book. Page 1 goes on the front!

# Explain a hook clue in your words

How Velcro Works · Curiosity day

Name _____ Date _____

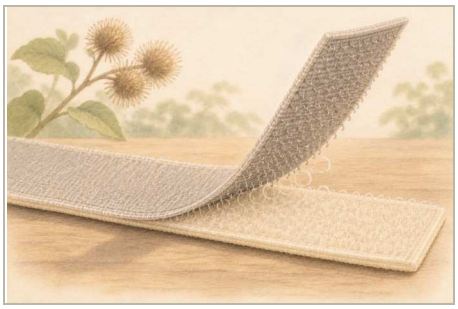
Choose one picture. Say your idea first, then draw or write it.



A burr on a walk



Hooks meet loops



Pull or peel?

Draw one hook and one loop. Show the hook catching the loop.

hook · loop · burr · peel · because · maybe

The idea came from a burr because … what was on its spines?

The strips hold together because … what do the hooks catch?

A peel opens easily because … how many hooks let go at once?

Does your sentence tell what you noticed and use because or maybe to explain?



# A Thousand Tiny Hands

Little Wonderer opened a strap on a shoe,
and it went RRRIP! like a grumpy kazoo.
"Who's hiding in there?" asked the small one, aghast.
"A thousand tiny hands," said the big one, "holding fast."

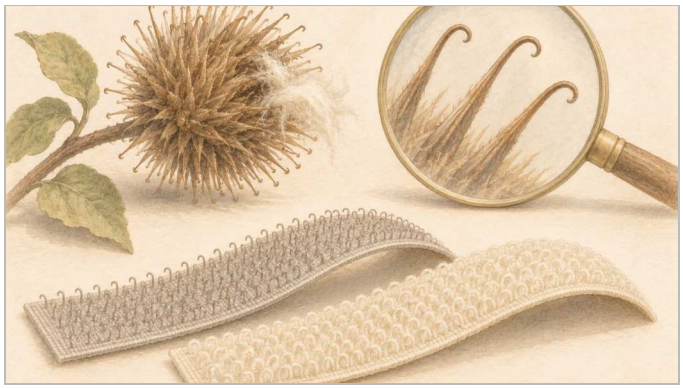
One side is all hooks, like a crowd of bent canes,
or umbrellas turned upside down, left out in the rains.
The other side's loops, like a soft curly mat,
and the hooks grab the loops, and that's simply that.

Big Wonderer said, "Slide them sideways — they'll hold,
every hook pulling hard, all together and bold.
But peel from a corner, a few at a time,
and they let go as easy as saying this rhyme."

Who dreamed it up first? Not a wizard or king,
but a burr on a dog, such a scratchy small thing.
So the little one bowed to a burr on its stem:
"Thank you for holding on. We learned it from them."

# How Velcro Works

## 01 Where did the idea come from?



A burr's spines end in tiny hooks.

**Look closer.** Find the tiny hooks at the tips of the burr's spines.

**A little explanation.** A burr. In 1941, a Swiss engineer came home from a walk with burdock burrs stuck to his socks and his dog's fur. Under a microscope, each spine ended in a tiny hook. Burrs cling on purpose: animals carry their seeds to new places.

**Try this.** After a walk outdoors, look closely at your socks. Did any seeds hitch a ride? Look at one with a magnifying glass for hooks.

## 02 How do you make a tiny hook?



Stiff hooks above, soft loops below.

**Look closer.** Compare the stiff hooks on top with the soft loops below.

**A little explanation.** You start with a loop. The first hooks were cotton and wore out. Nylon worked better: it was woven into loops, heated so the loops kept their shape, then a cutter snipped one side of each loop to leave a hook.

**Try this.** Bend a pipe cleaner into a small loop. Open one side a little so it becomes a hook. Can your hook catch a loop of yarn?

## 03 Why does a peel open so easily?



A peel opens just one thin line at a time.

**Look closer.** Look at how thin the line is where the strip is letting go.

**A little explanation.** A sideways pull, called shear, makes every hook hold at once. A peel opens only one thin line of hooks at a time. The rrrip is hooks springing free, one row after another.

**Try this.** Press two strips together. Slide them apart sideways, then peel one corner. Which one takes less push? Listen to each.

## 04 Why do old strips lose their grip?



Lint clogs the hooks; a comb lifts it out.

**Look closer.** Find the fuzz in the hooks and the clean part next to it.

**A little explanation.** Lint and fluff pack into the hooks, so they cannot reach the loops. A fine comb can lift the fuzz out, and the grip comes back.

**Try this.** Find a fuzzy strip on an old shoe or bag. Comb it gently with an old comb or toothbrush, then test the grip again.